In TEM, sample sections are ultrathin about 50–100 nm thick. These sections are placed on a copper grid and exposed to electron dense materials like lead acetate, uranylacetate, phosphotungstate. In SEM, samples can be directly imaged by mounting them on an aluminum stub.

Electron–Sample Interactions

Interaction of electron beam with the sample results in different types of electrons: Elastic scattered electrons, Inelastic scattered electrons, secondary electrons and backscattered electrons. Almost all types of electron interactions can be used to retrieve information about the specimen. Depending on the kind of radiation or emitted electrons which are used for detection, different properties of the specimen such as topography, elemental composition can be concluded. The following Figure 2.8 shows the interaction of the electron beam with the specimen.

In Transmission electron microscope (TEM), a beam of electrons is transmitted through an ultrathin specimen, interacting with the specimen as it passes through. An image is formed from the interaction of the transmitted unscattered electrons through the specimen.

Secondary electrons are mainly used in scanning electron microscope (SEM) for imaging the surface topography of biological specimens. The interaction of electron beam with samples results in secondary electrons and backscattered electrons that are detected by standard SEM equipment.

2.3.2 Working Principle and Instrumentation of TEM

The optics of the TEM is similar to conventional transmission light microscope. A transmission electron microscope has the following components,

1. Electron gun
2. Condenser lens
3. Specimen stage
4. Objective lens and projector lens
5. Screen/photographic film/Charged Coupled Device (CCD) camera

Electron Gun consists of a tungsten filament or cathode that emits electrons on receiving high voltage electric current (50,000–100,000 volts). A high voltage between the electron source (cathode) and an anode plate is applied leading to an electrostatic field through which the electrons are accelerated.

The emitted electrons travel through vacuum in the microscope column. Vacuum is essential to prevent strong scattering of electrons by gases. Electromagnetic condenser lenses focus the electrons into a very thin beam. Electron beam then travels through the specimen and then through the electromagnetic objective lenses. In a TEM microscope, the sample is located in the middle of the column. At the bottom of the microscope, unscattered electrons hit the fluorescent screen giving image of specimen with its different parts displayed in varied darkness, according to their density. The image can be studied directly, photographed or digitally recorded. Figure 2.9 show the arrangement

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